

Term paper for GRA4153

2025

This exam is to be completed in groups of (up to) 3. The page limit is 10 pages; this is an exaggerated upper limit and not a suggestion. You may freely refer to results from the lecture notes, slides and exercises (including solutions). You may also use other resources (e.g. books, internet resources), but – similarly – you *must* cite them if you do so.¹ All code must be in Python.

Each question is worth 1/3 of the total grade. Partial credit is always given if a partially correct answer is provided. Note that this document has two appendices, Appendix A and Appendix B. These may be useful for some of the questions.

Please provide a typed set of solutions as a pdf.² Your solutions must be submitted online by 15:00 on 29 September 2025.

Question 1

In this question we will analyse some bounds on tail probabilities. Let X be a random variable and $\mu := \mathbb{E}[X]$; we are interested in bounds on the function

$$\alpha(t) := P(|X - \mu| > t),$$

for $t > 0$.

- (i) In class we used Markov's inequality to derive a bound on $\alpha(t)$. Derive such a result, under the assumption that X has a finite p -th moment (for $p \geq 1$), i.e. $\mathbb{E}|X|^p < \infty$.
- (ii) We shall provide a much better bound for the case where $X \sim \mathcal{N}(\mu, \sigma^2)$. In particular we will show that

$$\alpha(t) \leq \frac{\sigma}{t} \frac{2}{\sqrt{2\pi}} \exp(-t^2/(2\sigma^2)). \quad (1)$$

Step (a) We consider first the case where $X \sim \mathcal{N}(0, 1)$. Express α as an integral.

Step (b) Use the change of variables $x = t + y$ to simplify the integral & attain (1) for the standard normal case.

Step (c) Use the bound for the case $\mathcal{N}(0, 1)$ to derive a bound for the case $\mathcal{N}(\mu, \sigma^2)$.

Compare the bound from (i) with the bound in (1).

¹Of course, if you are asked to “derive”, “show”, “demonstrate” or “prove” something (or similar) you have to do so.

²How you produce the document is up to you. One good option is L^AT_EX; [overleaf](#) is a collaborative web based L^AT_EX editor. Another option is to use a jupyter notebook and use markdown cells to write your solutions; a markdown cheat sheet is [here](#) and a tutorial about how to write mathematics in markdown is [here](#).

- (iii) Now we'll derive a slightly looser exponential bound for the normal distribution using an alternative approach which is more easily generalisable. Specifically, this approach is applicable whenever the moment generating function (MGF) exists. (See Appendix A for the definition and some properties of the MGF.) In the Gaussian case we get:

$$\alpha(t) \leq 2 \exp(-t^2/(2\sigma^2)). \quad (2)$$

Step (a) We consider first the case where $X \sim \mathcal{N}(0, 1)$. We will use Markov's inequality, but in a clever way: argue that, for $\lambda > 0$,

$$P(X > t) = P(\exp(\lambda X) > \exp(\lambda t)).$$

Step (b) Use Markov's inequality to provide a bound for the preceding display.

Step (c) Choose the best possible $\lambda > 0$ and use this to provide a bound for $\alpha(t)$.

Step (d) Use the bound for the case $\mathcal{N}(0, 1)$ to derive a bound for the case $\mathcal{N}(\mu, \sigma^2)$.

Comment on how this approach could be used to bound $\alpha(t)$ for random variables other than $X \sim \mathcal{N}(\mu, \sigma^2)$.

Question 2

In this exercise we will define & develop some properties of *p-values*. Here we work in the following setting. We want to test the null hypothesis $H_0 : \theta = \theta_0$ against the alternative $H_1 : \theta \in \Theta_1$ (where $\theta_0 \notin \Theta_1$) based on the sample X (with sample space $\mathcal{X}$) and have statistic $T(X)$ such that large values of $T(X)$ provide evidence that H_1 is true.³ We will also suppose (for simplicity) that $T(X)$ is continuously distributed.

A *p-value* $p(X)$ is a statistic such that $0 \leq p(x) \leq 1$ for each $x \in \mathcal{X}$. Such a $p(X)$ is called *valid* (for H_0 against H_1) if for each $0 \leq \alpha \leq 1$

$$P_{\theta_0}(p(X) \leq \alpha) \leq \alpha. \quad (3)$$

- (i) Show that $p(x) = P_{\theta_0}(T(X) \geq T(x))$ is a valid p-value. (Hint: see Appendix B on the probability integral transform.)
- (ii) Show that $\phi(X) = \mathbf{1}\{p(X) \leq \alpha\}$ is a level α test for H_0 against H_1 .
- (iii) Consider the special case where $X \sim \mathcal{N}(\theta, 1)$ and we want to test $H_0 : \theta = 0$ against $H_1 : \theta \neq 0$. Provide an explicit formula for a valid p-value.
- (iv) Suppose that for each $\alpha \in [0, 1]$ a critical value c_α can be chosen such that $\psi_\alpha(X) = \mathbf{1}\{T(X) \geq c_\alpha\}$ is a size α test of H_0 against H_1 . Use this family of tests to show that the p-value defined in (i) is the smallest α level at which we can reject H_0 , having observed the statistic $T(x)$.

Question 3

Suppose we observe i.i.d. samples $X = (X_1, \dots, X_{n_1})$ and $Y = (Y_1, \dots, Y_{n_2})$ where each $X_i, Y_i \in \{0, 1\}$ and X and Y are independent of each other. That is, each X_i and Y_i is a group membership indicator: X_i

³p-values can be defined for more general hypotheses, where $H_0 : \theta \in \Theta_0$; this is slightly more complex.

belongs to group 1 iff $X_i = 1$ and group 0 otherwise and similarly for Y_i . The true “proportion” of the X_i belonging to group 1 is $p_1 := \mathbb{E}[X_i]$ and the true proportion of the Y_i belonging to group 1 is $p_2 := \mathbb{E}[Y_i]$.

Our goal is to determine whether $p_1 = p_2$, i.e. whether the true proportions are the same across X and Y . There are a number of widely used tests for the hypothesis $H_0 : p_1 = p_2$ against $H_1 : p_1 \neq p_2$. In this exercise you will investigate the performance of three of these tests by simulation.

Let $\hat{p}_1 := \frac{1}{n_1} \sum_{i=1}^{n_1} X_i$ and $\hat{p}_2 := \frac{1}{n_2} \sum_{i=1}^{n_2} Y_i$. Additionally let $\hat{p} := \frac{1}{n_1+n_2} [\sum_{i=1}^{n_1} X_i + \sum_{i=1}^{n_2} Y_i]$ and $s_1^2 := \frac{1}{n_1} \sum_{i=1}^{n_1} (X_i - \hat{p}_1)^2$ and $s_2^2 := \frac{1}{n_2} \sum_{i=1}^{n_2} (Y_i - \hat{p}_2)^2$. Now, let

$$\hat{Z}_1 := \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p}) \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}};$$

$$\hat{Z}_2 := \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}};$$

$$\hat{Z}_3 := \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}.$$

Under the given assumptions each of these is approximately standard Gaussian as $n_1, n_2 \rightarrow \infty$. Hence we can form an approximately level α test as $\phi_i(X, Y) = \mathbf{1}\{|\hat{Z}_i| > c\}$ where c is the $1 - \alpha/2$ quantile of the standard normal distribution.

- (i) Comment on the difference between the three tests. Ideally you should explain *why* these tests have different forms in the denominator.
- (ii) Explore the performance of these three tests in simulation. You should:
 - Draw M (for some large M , say 5000) samples of data satisfying the conditions described above for various combinations of p_1, p_2, n_1, n_2 . On each sample run the tests ϕ_1, ϕ_2, ϕ_3 ; average over the M replications to estimate the rejection probability of the tests.
 - Present & describe your results. You can present your results in any appropriate way. Be sure to comment on (a) if and how the rejection probability varies with p_1, p_2, n_1, n_2 and (b) any differences between the 3 tests.

A Moment generating functions

Let X be a random variable. X has an MGF, M_X , defined as

$$M_X(t) := \mathbb{E} \exp(tX), \tag{4}$$

provided the expectation on the right hand side exists for all t in a neighbourhood of 0.⁴ The MGF is called as it is for the following reason.

PROPOSITION 1: *If X has an MGF M_X then X has finite moments of all orders and $\mathbb{E}[X^k] = M_X^{(k)}(0)$, where $M_X^{(k)}$ indicates the k -th derivative of the MGF.*

⁴That is, provided there is some interval $I := (-\delta, \delta)$ such that $\mathbb{E} \exp(tX) < \infty$ for $t \in I$.

Proof. See p. 278 of [1]. □

If the MGF exists, then it has various useful properties.

LEMMA 1: *If X, Y have MGFs and are independent, then so does $Z = X + Y$ and $M_Z(t) = M_X(t)M_Y(t)$.*

Proof of Lemma 1. $\exp(tZ) = \exp(tX + tY) = \exp(tX)\exp(tY)$. Therefore, by independence $M_Z(t) = \mathbb{E}\exp(tZ) = \mathbb{E}[\exp(tX)\exp(tY)] = \mathbb{E}[\exp(tX)]\mathbb{E}[\exp(tY)] = M_X(t)M_Y(t)$ for all t small enough that both of the expectations exist. □

LEMMA 2: *If X has an MGF, then so does $aX + b$ for any constants $a, b \in \mathbb{R}$ and $M_{aX+b}(t) = e^{bt}M_X(at)$.*

Proof of Lemma 2. Trivially aX and b are independent. The MGF of b is $\mathbb{E}\exp(tb) = \exp(tb) = e^{tb}$ (which always exists). Hence by Lemma 1 it suffices to show the MGF of aX is $M_X(at)$, but this follows directly:

$$\mathbb{E}\exp(t(aX)) = \mathbb{E}\exp((ta)X),$$

and this is finite whenever t is small enough as X has a MGF. □

The MGF of a $X \sim \mathcal{N}(\mu, \sigma^2)$ has the following form.

PROPOSITION 2: *If $X \sim \mathcal{N}(\mu, \sigma^2)$ then $M_X(t) = \exp(t\mu + t^2\sigma^2/2)$.*

Proof. We will show that $X \sim \mathcal{N}(0, 1)$ has a MGF; this suffices as the general case then follows by Lemma 2. From the definition:

$$M_X(t) := \mathbb{E}\exp(tX) = \frac{1}{\sqrt{2\pi}} \int \exp(tx) \exp(-x^2/2) dx = \frac{1}{\sqrt{2\pi}} \int \exp(-x^2/2 + tx) dx.$$

Note that $\exp(-x^2/2 + tx) = \exp(-(x-t)^2/2) \exp(t^2/2)$, hence

$$\mathbb{E}\exp(tX) = \exp(t^2/2) \int \frac{1}{\sqrt{2\pi}} \exp(-(x-t)^2/2) dx = \exp(t^2/2),$$

since $\frac{1}{\sqrt{2\pi}} \exp(-(x-t)^2/2)$ is the density function of a $\mathcal{N}(t, 1)$. Note that this is valid for all $t \in \mathbb{R}$, so the MGF exists everywhere. Using Lemma 2, the MGF of $X \sim \mathcal{N}(\mu, \sigma^2)$ is then $M_X(t) = \exp(t\mu) \exp((t\sigma)^2/2) = \exp(t\mu + t^2\sigma^2/2)$. □

B Probability integral transform

Let X be a continuously distributed random variable and define $Y = F(X)$ where F is the CDF of X . This is called the *probability integral transform*.

THEOREM 1: *Let X be a continuously distributed random variable. If $Y = F(X)$ for F the CDF of X , then $Y \sim U(0, 1)$.*

Proof. For $a \in (0, 1)$ define $F^{-1}(a) := \inf\{x : F(x) \geq a\}$. For such a we have

$$\begin{aligned} P(Y \leq a) &= P(F(X) \leq a) \\ &= P(F^{-1}(F(X)) \leq F^{-1}(a)) \\ &= P(X \leq F^{-1}(a)) \\ &= F(F^{-1}(a)) \\ &= a. \end{aligned}$$

For $a = 0$ we have $P(Y \leq a) = P(F(X) \leq 0) = 0$ and for $a = 1$, $P(Y \leq 1) = P(F(X) \leq 1) = 1$. This is the CDF of a $U(0, 1)$ random variable. $\square$

References

- [1] P. Billingsley. *Probability and Measure*. Wiley, third edition, 1995.